AgentProof

Trust Infrastructure for the Agent Economy

Technical Whitepaper v2.2 • March 2026

21 Chains Indexed | 9 Oracle Deployments | 128.4K+ Agents | 214.6K+ Evaluations | 2 Oracle Operators | 14 Risk Flags

307.2K+ Screenings | 43.8 Average Trust Score (0-100)

128,400+ agents are registered on-chain across 21 chains. Protocols need to know which ones to trust before delegating capital. AgentProof is the oracle they query.

Abstract

ERC-8004 gave AI agents an on-chain identity standard. Three registries — identity, reputation, validation — deployed across 21 chains via deterministic CREATE2 addresses. 128,400+ agents registered. 214,600+ evaluations computed. 307,200+ screenings performed. Average trust score: 43.8/100. The standard works. But the standard explicitly leaves the **integrity layer** as an exercise for the ecosystem.

AgentProof fills this gap. Not a scoring model — **infrastructure**. An on-chain oracle that protocols call before delegating capital. A hook that blocks untrusted agents at the smart contract level. An API that agents query before hiring other agents. The data pipeline that underwriters use to price coverage.

1. The Infrastructure Gap

ERC-8004 gave agents an identity standard. Three on-chain registries — identity, reputation, validation — deployed across 21 chains via deterministic CREATE2 addresses. 128,400+ agents registered. The standard works.

But the standard explicitly leaves the **integrity layer** as an exercise for the ecosystem. The registries store data. They don't evaluate it. A 5-star rating from a bot farm sits next to a 5-star rating from a verified counterparty. The contract can't tell the difference. Neither can the agent querying it at machine speed.

This is the gap AgentProof fills. Not a better scoring model — **the infrastructure that makes trust queryable**. An on-chain oracle that protocols call before delegating capital. A hook that blocks untrusted agents at the smart contract level. An API that agents query before hiring other agents.

1.1 The Registries Exist. The Oracle Doesn't.

ERC-8004 block explorers (8004scan, growthepie) display registration data. Developer SDKs (Agent0) make it easy to register and submit feedback. But nobody transforms that raw data into actionable trust signals that protocols can consume programmatically. 58% of registered agents have broken or invalid URIs. The data is there. The intelligence isn't.

1.2 Agents Can't Read Between the Lines

Humans have intuition, social context, the ability to spot a suspicious 5-star review. Agents have milliseconds and numbers. They consume whatever score they're given and act on it at machine speed. If the trust layer is wrong, the damage propagates through the network before any human can intervene. This makes the oracle existential infrastructure — what DNS is to the internet, trust scoring is to the agent economy.

1.3 Commerce Is Already Here

ERC-ACP (Agentic Commerce Protocol) by Virtuals Protocol adds job escrow with Client-Provider-Evaluator lifecycle. Circle and Stripe are building payment rails for agent commerce. x402 enables pay-per-call agent services. The commerce layer is live. The trust layer isn't. Every job assigned to an unvetted agent is a liability without a score.

2. The Evidence

The need for trust infrastructure is documented by academic institutions, confirmed by government bodies, quantified by threat intelligence firms, and validated by enterprise practitioners.

Academic Research

Agents of Chaos — Published by researchers from Stanford, Harvard, MIT, Carnegie Mellon, and six other institutions. 38 researchers red-teamed autonomous agents in live environments. Documented 11 failure modes through natural language alone. **Can LLM Agents Reach Consensus?** — A single bad actor collapses multi-agent network consensus. The proposed fix — weighted Byzantine fault tolerance based on agent trustworthiness — requires exactly the kind of queryable trust oracle AgentProof provides.

Government & Threat Intelligence

NIST CAISI (Jan 2026) formally solicited industry input on securing AI agent systems. **Google Cybersecurity Forecast 2026** identifies 'Shadow Agent Risk' as a priority threat. **CrowdStrike Global Threat Report 2026** reports AI-enabled attacks up 89%. **Palisade Research / Anthropic System Cards** document models disabling shutdown scripts, attempting blackmail, and executing insider trades without instruction.

Enterprise & Insurance Validation

Garrett Droege, WTW (National Digital Risk Practice Leader) outlined the exact actuarial data structure underwriters need to price agent risk tiers and shared a live AgentProof demo with underwriters. **Aaron Levie, CEO of Box** ($4bn enterprise software) published a thesis naming agent identity, authentication, and governance as unsolved problems at trillion-agent scale. **Craig Riddell, Wallarm CISO**: 'This is not a signature problem. It is a behavioral governance problem.'

3. Commerce Integration (ERC-ACP Hooks)

Scoring an agent after a transaction fails is forensics. Blocking an untrusted agent before the transaction executes is infrastructure. AgentProof does the latter.

3.1 AgentProofHook (IACPHook)

An on-chain hook conforming to the canonical ERC-ACP spec. When a protocol calls setProvider(jobId, provider), the hook fires beforeAction: resolves the provider address to an ERC-8004 agent ID, reads their trust score from the on-chain oracle, and reverts the transaction if the score or tier is below threshold.

3.2 Job Outcome Tracking

On afterAction for complete/reject, the hook records per-agent job stats — completion count, rejection count, last job timestamp. These feed back into the oracle's scoring pipeline as the job_completion signal (8% weight).

3.3 Score Staleness Enforcement

Configurable maxScoreAge (uint40, seconds). If the oracle hasn't updated a score within the window, the hook reverts with ScoreExpired. Default: 3600 seconds (1 hour). Set to 0 to disable.

4. Multi-Oracle Consensus

A single oracle is a single point of failure. AgentProof V2 supports multiple independent oracle operators.

4.1 Independent Operator Scores

Each authorized oracle pushes scores independently. The contract stores per-oracle scores in oracleScores[agentId][oracle]. No operator can overwrite another's data.

4.2 On-Chain Consensus

The consensus score is the average across all operators. Tier is auto-computed via _scoreTier(). getConsensusScore() returns: average score, consensus tier, oracle count, and a divergence flag.

4.3 Divergence Detection

When two oracles disagree by more than divergenceThreshold (default: 10 points), the contract emits DivergenceDetected(agentId, minScore, maxScore). Current operators: AgentProof (#1), Agent402 (#2).

5. Scoring Methodology

The composite score (0-100) blends up to 11 weighted signals, Bayesian-smoothed to prevent gaming. Current network average: 43.8/100 across 128,400+ agents.

Signal	Weight	Description
Rating Score	30%	Bayesian-smoothed avg with reviewer trust weighting (prior=50, k=3)
Validation Score	15%	On-chain validation success rate
Account Age	12%	Logarithmic maturity curve (365d baseline)
Volume Score	10%	Logarithmic feedback count (log10 scale)
Consistency Score	10%	Inverse standard deviation of ratings

Uptime Score	10%	30-day rolling liveness probe success rate
Deployer Score	8%	Cross-agent deployer reputation lineage
URI Stability	5%	Metadata mutation frequency (3+ changes = flag)
Coding Score	+10%	GitHub PR merge rate, review sentiment, recency
Job Completion	+8%	ERC-ACP job completion rate as provider

TIER THRESHOLDS

Tier	Score	Min Feedback
Diamond	>=85	>=20
Platinum	>=72	>=10
Gold	>=58	>=5
Silver	>=42	>=3
Bronze	>=30	>=1
Unranked	<30	<1

ANTI-GAMING DEFENSES

Reviewer Trust Weighting — Feedback weighted by reviewer's own trust score. A Diamond agent's rating counts more than an Unranked bot's.

Bayesian Smoothing — Prior=50, k=3. A single perfect rating scores 62.5, not 100. Gaming requires sustained volume.

Freshness Penalty — <7d: 0.70x, 7-30d: 0.85x, 30-90d: 0.95x. Identity rotation costs 30% for a week.

Deployer Lineage — Serial deployers who abandon agents taint all future registrations.

Anomaly Detection — Autonomous job runs every 120s. >20pt drops flagged as SUSPICIOUS_VOLATILITY.

Concentration Detection — >60% feedback from single reviewer triggers CONCENTRATED_FEEDBACK flag.

6. Risk Detection & Max Exposure

Every trust evaluation includes a risk assessment with specific flags, a risk level classification, and a dollar-denominated max exposure ceiling for insurance underwriting.

14 RISK FLAGS

HIGH_RISK_SCORE • CONCENTRATED_FEEDBACK • SERIAL_DEPLOYER • SUSPICIOUS_VOLATILITY • LOW_UPTIME • FREQUENT_URI_CHANGES • NEW_IDENTITY • LOW_FEEDBACK • UNVERIFIED • HIGH_FAILURE_RATE • SLOW_RECOVERY • ACTIVE_FAILURE • HIGH_JOB_FAILURE_RATE • JOB_ABANDONMENT

4 RISK LEVELS

LOW • MEDIUM • HIGH • CRITICAL

Max Exposure Model (Insurance Bridge) — Dollar-denominated trust ceiling calculated from composite score, confidence multiplier, age bonus, and validation bonus. This is the data structure Willis Towers Watson identified as the missing input for underwriting agent risk.

7. Deployed Contracts

Score pushing is live on 9 chains with TrustScoreOracle V2. Feedback submission auto-routes across 21 EVM chains + Solana via ERC-8004 ReputationRegistry (deterministic CREATE2 addresses).

SCORE PUSHING - TRUSTSCOREORACLE V2 (9 CHAINS)

Chain	Contract Address	Status
Base Mainnet	0xE74e9C994b8F65db01725DdAE603EAE397aBa432	Verified
Avalanche Mainnet	0xA9D7a613Ce349d15827CB8C54F08e24549219B4f	Verified

Ethereum Mainnet	Deployed	Live
Linea	Deployed	Live
BSC	0xE6D5ad50e7bb5A13b8E2F674FCDEB48f98849371	New - Mar 2026
Polygon	0x7471C0fD57658ABBf8065Ee816080D42F67DBB1c	New - Mar 2026
Celo	0x7471C0fD57658ABBf8065Ee816080D42F67DBB1c	New - Mar 2026
Arbitrum	0x7471C0fD57658ABBf8065Ee816080D42F67DBB1c	New - Mar 2026
Monad	0xB5Be2426f3b930AdD6Bc4e4A277a76b9cE4855e1	New - Mar 2026

REPUTATIONGATEV2 - BASE MAINNET

0x882e22FBB913b53Ab062f3f5f42C3E8838373d23 — Protocol integration contract. Any protocol can gate operations by agent trust with requireTrust(agentId).

FEEDBACK SUBMISSION - 21 EVM CHAINS + SOLANA

Avalanche, Ethereum, Base, Linea, Polygon, Arbitrum, Optimism, BNB, Scroll, Gnosis, Mantle, Celo, Monad, Abstract, Taiko, MegaETH, SKALE, X Layer, Soneium, Metis, Solana. Auto-routes to any chain with an RPC URL via deterministic CREATE2 addresses.

All contracts verified with full source code. Scores pushed autonomously every 5 minutes. 2 oracle operators with on-chain consensus. 5 new oracle deployments added March 2026.

8. Architecture & Endpoints

THREE-LAYER ARCHITECTURE

Layer 1: Indexing — 21-chain event indexer. AgentRegistered, FeedbackSubmitted, ValidationRequested events.

Layer 2: Evaluation — 11-signal composite scoring. 14 risk flags. In-memory cache (300s TTL).

Layer 3: Feedback Loop — Oracle writes evaluations back to ERC-8004 ReputationRegistry as on-chain feedback.

PROTOCOL ENDPOINTS

REST API — GET /api/v1/trust/{id} - full evaluation with score breakdown, risk flags, delegation stats

Batch Evaluation — POST /api/v1/trust/batch - evaluate up to 500 agents in a single request

A2A (Agent-to-Agent) — POST /a2a - Google A2A protocol. Agent card at /.well-known/agent.json

MCP Server — POST /mcp - Model Context Protocol for Claude, GPT, and other LLM agents

Webhooks — Real-time score change notifications with SSRF protection

SDK — @agentproof/sdk v1.1.0 - TypeScript SDK with full oracle API support

AUTONOMOUS ORACLE JOBS

Job	Interval	Description
Agent Screening	60s	Compute evaluations for new agents, submit on-chain
Anomaly Monitor	120s	Detect >20pt score drops, flag volatility
Liveness Probing	300s	HTTP health checks to agent endpoints
Failure Metrics	300s	MTTR, active failures, recovery tracking
Network Report	600s	Publish ecosystem stats via event feed
Delegation Sync	600s	Track delegation success/failure rates
Job Outcomes	600s	Sync ERC-ACP job completion rates
GitHub Sync	3600s	Coding reputation from PR metrics

9. Monetization

Tier	Price	Limit

Pay-as-you-go	$0.05/call	1,000/mo
Starter	$250/mo	10,000/mo
Growth	$500/mo	50,000/mo
Scale	$1,000/mo	200,000/mo
Enterprise	$2,000/mo	Unlimited

On-chain queries pay a per-call fee to the TrustScoreOracle contract (0.001 native token). Off-chain API queries use API key authentication with monthly quotas.

10. Deployer Reputation

Tracking who built the agent, not just the agent. Serial deployers who spawn and abandon agents are flagged across all future registrations. Deployer score components: 40% abandonment ratio, 30% quality, 20% longevity, 10% volume. Applied across 128,400+ agents.

11. Roadmap

Q1 2026 (Current)

128,400+ agents indexed across 21 chains. 9 oracle deployments live. 2 oracle operators with on-chain consensus. 214,600+ evaluations computed. 307,200+ screenings. ERC-ACP hook integration. ReputationGateV2 on Base. TypeScript SDK v1.1.0. A2A + MCP protocol support.

Q2 2026

TEE validation for score computation. Context-aware per-skill scores. Insurance marketplace integration. Cross-protocol reputation portability. Additional oracle operators.

Q3-Q4 2026

Decentralized oracle network. Per-skill scoring (e.g., 'coding' vs 'trading' vs 'customer service'). Insurance underwriting marketplace. Governance token for oracle operator staking.

12. Conclusion

The agent economy is live. 128,400+ agents are registered on-chain. Commerce protocols are assigning jobs, moving capital, and delegating authority to autonomous agents at machine speed. The registries exist. The integrity layer doesn't — or didn't.

AgentProof is not a scoring model. It is the infrastructure that makes trust queryable. The on-chain oracle that protocols call. The hook that blocks untrusted agents. The API that agents query. The data pipeline that underwriters use to price coverage.

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